

Brian Woods

Staff / Principal ML & Systems Engineer

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SUMMARY

Engineer with 30 years in production systems, now working at the intersection of ML research and systems engineering. I build and operate a live ML trading platform on Solana, publish causal research on blockchain infrastructure, and am an active open-source contributor to vLLM and Pydantic AI. Nine startups as founder, CTO, or director; Yahoo Mail's MTA before that; a 2026 Berkeley M.S. in data science to put formal foundations under the practice. I report true numbers, including my own.

OPEN SOURCE

Active contributor to vLLM and Pydantic AI: five open PRs and a review contribution across the two projects.

- **vLLM** (LLM inference engine): authored fixes for silently ignored configuration (#52557, #52546); review contributor on KV cache correctness (#52530), surfacing a decode-path livelock that became the fix's second commit. Analysis published at brian.biz.
- **Pydantic AI** (agent framework): filed and fixed correctness issues in the embeddings and AG-UI streaming surfaces (#7508, #7520, #7521).

EXPERIENCE

Independent ML Engineer: Real-Time Trading Systems & Published Research (self-employed) · 2024 – Present

- Built and operate an autonomous ML trading platform on Solana: gRPC/Geyser streaming (15M+ transactions per run), multi-region Jito execution, Rust/Anchor programs with Python orchestration and Rust extension modules on the hot path. Live public telemetry at brian.biz/leeroy; P&L flat to negative to date, reported plainly.
- RL systems in production: PPO and behavioral cloning (Memeorator), offline CQL, stat-arb models (68.5% win rate, 3.94 profit factor on paper trading). Full MLOps built solo: training, drift monitoring, champion/challenger, kill rules.
- Published causal research on blockchain propagation (brian.biz/optimum): Ethereum RD study on 216k mainnet slots; live Solana validator study with fixed effects and wild cluster bootstrap, re-estimated daily.
- Agent benchmark reliability study (brian.biz/deccan) and audio data-quality ablation (brian.biz/david.ai, pipeline complete). Open-source human-gated agentic pipeline at github.com/brianosaurus/agentic-workflow.
- Built concurrently with the UC Berkeley M.S. (2024–2026).

AGEN.AI · CTO (bootstrapped, never raised funding) · [dates]

- Built the company's first website and backend personally; set architecture and engineering practice.

Umee · Protocol Engineer, Go (company wound down) · [dates]

- Built IBC (inter-blockchain communication) frameworks on the Cosmos SDK.

Sifchain · Protocol Engineer, Go (company wound down) · [dates]

- IBC framework work on a Cosmos SDK chain.

Augur · Senior Engineer · [dates, ~3 years]

- Decentralized prediction market: venue-side work spanning market lifecycle, browser architecture, and event retrieval.

HelloBit · Director of Engineering · 2013 – 2015

- Bitcoin remittance. Recruited and led the engineering team.

AnythingSocial · Director of Engineering · [dates]

- Led development across distributed teams.

Bebo · Head of Growth Engineering · [dates]

- Growth as measurement: funnels, cohorts, experiments, at international scale.

Yahoo · Software Engineer, Yahoo Mail · 1997 – [end]

- Authored the Yahoo Mail MTA in C++, processing 6B+ messages/month. 6 U.S. patents.

EDUCATION

M.S., Information & Data Science (AI concentration), UC Berkeley · 2024 – 2026

- Capstone VitalFoods.ai (team project): designed the original RotatE knowledge-graph models and built the full site; a teammate led embedding training. Ran all SageMaker/EC2 infrastructure. Completed while building and operating the trading platform.

B.S., Computer Science & Economics, University of Colorado Boulder · 1996

PATENTS & CERTIFICATIONS

- 6 U.S. patents in bulk-email processing and anti-spam (Yahoo Mail).
- Cryptography I (Stanford) · GCP Big Data & ML Fundamentals.

SKILLS

ML research & evaluation: experiment design and pre-registration, causal inference (RDD, difference-in-differences, fixed-effects panels, placebo tests), benchmark and eval construction, verifiable-reward task design, bootstrap uncertainty and power analysis, honest reporting of negative results

ML engineering: PyTorch, Hugging Face transformers/datasets, RL training (PPO, CQL, behavioral cloning), LightGBM/XGBoost, PyMC, feature pipelines, drift monitoring, champion/challenger, AWS SageMaker and EC2 run hands-on, production MLOps built from scratch; LLM serving internals (vLLM, SGLang: continuous batching, paged KV cache, prefix caching)

Systems: real-time streaming (gRPC), multi-region execution, queueing and backpressure, fail-closed design, observability and public telemetry, resumable manifest-driven pipelines; Python, Rust (extension modules in the Python interpreter), Go, C++, SQL, TypeScript/JavaScript

Leadership & practice: hired and mentored across four leadership stints; standards for research rigor, documentation, and reproducibility; AI agents in the full engineering loop daily (spec, adversarial plan review, build, test); MCP server development